Width-independent generalization in the strong-prior limit of Bayesian neural networks

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Abstract

Can a Bayesian neural network learn a function with a small implementation even when the learner has arbitrarily many neurons? We prove a positive result for noiseless Boolean prediction under a layer-scaled Gaussian prior, conditioned to give a fixed output margin at every input. At each fixed depth, the number of examples needed for small posterior prediction error is polynomial in the input dimension and the target's bounded-weight network complexity, independently of the learner's width. This complexity jointly bounds the width of an implementing network and the magnitude of every weight. The learner can have any width large enough to contain the target; its prior variance must then be sufficiently small, with a threshold that may depend on that width. The key prior statement is that function probabilities, at their small-variance exponential rate, are governed by a complexity measure polynomially equivalent to bounded-weight network complexity. We prove this by combining a Gaussian rate calculation, a neuron-cloning construction, and compression by reverse-order subsampling. The resulting learning guarantee is polynomial at fixed depth, but does not give polynomial dependence on depth. A separate tanh construction shows an obstruction to obtaining such a depth bound by uniform norm-sublevel counting alone.

1 Introduction

Suppose a Boolean target function has a small neural-network implementation. Can a much wider Bayesian neural network learn it from a number of examples determined by the target's complexity, rather than by the learner's number of parameters? The learner can fit the same sample in many ways, so representability alone gives no answer. A positive result must explain why its prior favors functions simple enough to generalize.

Here Bayesian neural-network learning, or NNBayes, means conditioning a distribution on network parameters on exact agreement with noiseless training labels. We use fully connected networks with hidden activations satisfying $|\phi| \leq 1$ and $\mathrm{Lip}(\phi) \leq 1$, and a linear readout; tanh is a standard example. The parameter prior is Gaussian with different variances in different layers. Before training, we restrict it to networks whose output has absolute value greater than $1/2$ at every Boolean input. Each retained network therefore defines a Boolean function with a uniform output margin. Prediction uses the posterior distribution on these functions, and training and test inputs come from the same arbitrary distribution.

Our main theorem answers the question affirmatively in the *strong-prior limit*: the prior variance tends to zero at a fixed learner width. For every fixed depth L, a target of bounded-weight network

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complexity C can be learned from a number of examples polynomial in the input dimension d, in C, and in the inverse accuracy, and logarithmic in the inverse failure probability. This sample bound is independent of the learner width N, and holds for every $N \geq C$. The order matters: choose any such finite width, and then take the variance sufficiently small. The required variance threshold may depend on N; the theorem does not require or take an infinite-width limit. Also, "polynomial" here means at fixed depth: the bound need not be polynomial in L.

The argument rests on a statement about the prior itself. At small variance, the exponential rates of relative function probabilities are given by differences between the minimum weighted squared parameter norms of their margin-preserving implementations. We show that this minimum norm is polynomially equivalent to bounded-weight network complexity, uniformly over larger learner widths. Here the complexity is the least integer that bounds both the implementing network's width and all its weight magnitudes. We call the induced function prior a *circuit prior* in this sense: its exponential rates are polynomially related to the complexity of bounded-weight implementations. Boolean circuits of bounded fan-in with C gates and depth less than L also supply depth-L bounded-weight network implementations with $C_L(f) = O(C)$ when the activation is nonconstant, as we prove in section 8.

Two constructions establish the complexity comparison. Cloning neurons embeds a small target network into any larger width at bounded norm cost. Conversely, any network of bounded weighted norm can be compressed to polynomial width while preserving its Boolean function. Compression is the main technical step: it first removes large entries using thresholds proportional to their row norms, then samples hidden neurons in reverse layer order. Reverse order makes each concentration bound range over the already compressed next layer, rather than over the original width. Counting these compressed networks bounds the number of functions whose norm is no larger than the target's. All higher-norm functions have vanishing total prior mass relative to the target as the variance tends to zero, so an Occam argument yields the learning theorem. This comparison requires no absolute lower bound on the target's prior probability.

The paper develops the main proof thread of Vaintrob and Hänni's lecture notes [1], with the model, statements, and proofs given here in full. Section 2 states the learning theorem; the subsequent sections establish the Gaussian rate, cloning, compression, and counting steps. The final sections relate the result to Boolean circuits and discuss its depth dependence. An appendix gives a tanh example showing that a uniform count of low-norm functions can have severe depth dependence. That example concerns the proof method, rather than proving a sample-complexity lower bound for the Bayesian learner.

Relation to earlier work. The statistical step is a realizable Occam argument [2], adapted to a posterior with a small tail outside a finite set. Norm-based capacity control independent of parameter count has a long history, including Bartlett [3], Neyshabur, Tomioka, and Srebro [4], and Golowich, Rakhlin, and Shamir [5]. Their activation and norm assumptions differ, so we give a self-contained compression and counting proof. Buzaglo et al. [6] also analyze generalization of overparameterized networks through probabilities of fitting a teacher under random parameter sampling. Here the organizing quantity is a relative small-variance rate, and the theorem's width independence applies to sample complexity rather than to the required variance threshold.

2 Setting and main result

Fix an input dimension $d \geq 1$ and depth $L \geq 2$. Write $\mathcal{X} = \{-1, 1\}^d$ and $\bar{x} = (1, x_1, \ldots, x_d)$. All $L - 1$ hidden layers have width N. For a fixed activation $\phi : \mathbb{R} \to \mathbb{R}$ satisfying

$$|\phi(t)| \leq 1, \qquad |\phi(t) - \phi(s)| \leq |t - s|, \tag{1}$$

the network is

$$h^0(x) = \bar{x}, \qquad h^\ell(x) = \phi(W^\ell h^{\ell-1}(x)) \quad (1 \leq \ell < L), \qquad z_w(x) = W^L h^{L-1}(x).$$

The activation acts coordinatewise, and the readout W^L is linear. There are no separate biases beyond the constant input coordinate; constant hidden neurons may carry an offset to later layers.

For a Boolean function $f : \mathcal{X} \to \{-1, 1\}$, let

$$U_{N,f} = \{w : f(x) z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}.$$

These sets are open, disjoint, and indexed by a finite set of functions. Let $\mathcal{H}_N = \{f : U_{N,f} \neq \varnothing\}$ and $U_N = \bigcup_{f \in \mathcal{H}_N} U_{N,f}$. We only use widths for which U_N is nonempty.

Definition 2.1 (Norm and target complexity)**.** The weighted squared norm of a width-N network is

$$\kappa_N(w) = \frac{d}{N} \left\|W^1\right\|_F^2 + \sum_{\ell=2}^{L-1} \left\|W^\ell\right\|_F^2 + N \left\|W^L\right\|_2^2, \qquad \kappa_N(f) = \inf_{w \in U_{N,f}} \kappa_N(w). \tag{2}$$

An empty sum is zero and an infimum over the empty set is $+\infty$. The bounded-weight network complexity $C_L(f)$ is the smallest integer $C \geq 1$ such that $U_{C,f}$ contains a network all of whose weights have absolute value at most C.

We assume $C_L(f) < \infty$ for the target. The learning theorem itself needs no non-polynomiality assumption beyond (1). To connect C_L to ordinary Boolean circuits, we additionally take ϕ to be nonconstant; boundedness then makes it non-polynomial. Tanh is a running example.

Under the unconditioned Gaussian law $\mu_{N,\sigma}$, all weights are independent, with standard deviations

$$\sigma_1 = \sqrt{N/d}\,\sigma, \qquad \sigma_\ell = \sigma \ (2 \leq \ell < L), \qquad \sigma_L = \sigma/\sqrt{N}. \tag{3}$$

Its density is proportional to $\exp[-\kappa_N(w)/(2\sigma^2)]$. Before observing data, condition this law on U_N. The induced function prior is

$$\pi_{N,\sigma}(g) = \frac{\mu_{N,\sigma}(U_{N,g})}{\mu_{N,\sigma}(U_N)}.$$

This conditioning restricts the prior at every input, including inputs that will not appear in the training sample. It is imposed before any labels are observed.

Let D be any distribution on $\mathcal{X}$. Given $S_n = ((x_i, f(x_i)))_{i=1}^n$ with independent $x_i \sim D$, the posterior $\Pi_{N,\sigma}(\cdot \mid S_n)$ is $\pi_{N,\sigma}$ restricted to functions consistent with all labels. Define

$$\mathrm{err}_D(g) = \mathbb{P}_{x \sim D}\{g(x) \neq f(x)\}, \qquad R_D(\Pi) = \mathbb{E}_{g \sim \Pi} \mathrm{err}_D(g).$$

Thus $1 - R_D(\Pi)$ is the average posterior probability of the correct label. A posterior-majority classifier, with any tie rule, has error at most $2R_D(\Pi)$.

Theorem 2.2 (Strong-prior generalization)**.** *For every fixed $L \geq 2$ there is a polynomial p_L with the following property. Let f have $C = C_L(f) < \infty$, and let $N \geq C$ be arbitrary. For every $\varepsilon \in (0,1)$ there is $\sigma_0 > 0$ such that, for all $0 < \sigma < \sigma_0$, all distributions D on $\mathcal{X}$, all $\delta \in (0,1)$, and all*

$$n \geq \frac{2}{\varepsilon}\left(p_L(d,C) + \log\frac{1}{\delta}\right), \tag{4}$$

with probability at least $1-\delta$ over S_n,

$$R_D(\Pi_{N,\sigma}(\cdot \mid S_n)) \leq \varepsilon.$$

The polynomial is independent of N and of ϕ. The threshold σ_0 may depend on $d, L, N, \phi, f, \varepsilon$, but not on D, n, or δ.

The theorem controls the posterior probability of the correct label, averaged over a fresh test input. In particular, choosing $\varepsilon < 10^{-3}$ gives average correct-label probability greater than 99.9%, and the posterior-majority classifier has error at most 2ε. Its dependence on confidence is logarithmic: one may take $\delta = e^{-C}$ and retain a polynomial sample bound. The order of quantifiers matters: we first fix a width, however large, and then take the variance sufficiently small. The theorem provides neither a rate for σ_0 nor a simultaneous $N \to \infty$, $\sigma \to 0$ theorem.

3 Relevant functions and the small-variance limit

Lemma 3.1 (A finite relevant set)**.** *Let π be a prior on a countable class containing a target f with $\pi(f) > 0$. Suppose a finite set A contains f, $|A| \leq M$, and*

$$\pi(\mathcal{H} \setminus A) \leq \tau\pi(f).$$

For any $a, \delta \in (0,1)$, if $n \geq a^{-1}(\log M + \log(1/\delta))$, then the consistency posterior satisfies $R_D(\Pi) \leq a + \tau$ with probability at least $1-\delta$.

Proof. A function of error greater than a survives n independent examples with probability at most e^{-an}. A union bound over A shows that, except on an event of probability at most Me^{-an}, every consistent member of A has error at most a. On every training sample, the posterior denominator is at least $\pi(f)$, so the posterior mass outside A is at most τ. Its contribution to the posterior risk is at most τ; the contribution inside A is at most a. □

This lemma does not require A to consist of the highest-prior functions. Nor does it require a unique simplest consistent function. Both points are useful when several functions have the same minimum norm.

Lemma 3.2 (Gaussian rate for a function)**.** *At fixed d, L, N, for every $g \in \mathcal{H}_N$,*

$$\lim_{\sigma \downarrow 0} 2\sigma^2 \log \mu_{N,\sigma}(U_{N,g}) = -\kappa_N(g). \tag{5}$$

Consequently, if $\kappa_N(g) > \kappa_N(f)$, then $\pi_{N,\sigma}(g)/\pi_{N,\sigma}(f) \to 0$.

Proof. Use coordinates $u^\ell = \sqrt{c_\ell}W^\ell$, where $c_1 = d/N$, $c_L = N$, and the other c_ℓ equal one. Then $\kappa_N(w) = \|u\|_2^2$ and $u \sim \mathcal{N}(0, \sigma^2 I_P)$, where P is the parameter count. Let V be the image of $U_{N,g}$ and $b = \inf_{u \in V} \|u\|^2$.

For $0 < \theta < 1$, the Gaussian exponential-moment bound gives

$$\mathbb{P}\{u \in V\} \le \mathbb{P}\{\|u\|^2 \ge b\} \le \theta^{-P/2} \exp\left(-\frac{(1-\theta)b}{2\sigma^2}\right).$$

Take the upper limit and then $\theta \downarrow 0$. Conversely, for any $r > 0$, openness and the definition of b give a positive-volume ball $B \subset V$ on which $\|u\|^2 < b + r$. Integrating the density over B gives

$$\mathbb{P}\{u \in V\} \ge \mathrm{vol}(B)(2\pi\sigma^2)^{-P/2} \exp\left(-\frac{b+r}{2\sigma^2}\right).$$

Take the lower limit and then $r \downarrow 0$. This proves (5). Ratios of the conditioned function probabilities equal ratios of the unconditioned event probabilities, so their logarithmic rate is $-\kappa_N(g)+\kappa_N(f)$. □

For $A_{N,f} = \{g \in \mathcal{H}_N : \kappa_N(g) \le \kappa_N(f)\}$, finiteness of $\mathcal{H}_N$ now implies

$$\frac{\pi_{N,\sigma}(\mathcal{H}_N \setminus A_{N,f})}{\pi_{N,\sigma}(f)} \longrightarrow 0. \tag{6}$$

Equal-cost functions all remain in $A_{N,f}$; no claim about their limiting probability ratios is needed. Since there are finitely many targets at fixed d, N, a single sufficiently small variance can also enforce any prescribed tail ratio for every representable target at that width. This observation supplies no quantitative or width-uniform threshold.

4 Cloning: a small target stays cheap

Lemma 4.1 (Cloning)**.** *If a width-C network implements f, then for every $N \ge C$ a width-N network implements the same real-valued output and has weighted norm at most $2\kappa_C(w)$. If N is a multiple of C, the norm is preserved exactly. In particular, for $C = C_L(f)$,*

$$\kappa_N(f) \le B_L(d, C) := d(d+1)C^2 + LC^4. \tag{7}$$

Proof. Set $k = \lfloor N/C \rfloor$ and replace each hidden neuron by k identical copies. Copy its first-layer incoming weights unchanged. For each later layer, divide every weight leaving a copied neuron by k; connect every destination copy in the same way. The copies compute identical activations, and their outgoing contributions sum to the original contribution. Pad unused positions with zero incident weights. This works even when $\phi(0) \neq 0$, since the padded neurons have no outgoing effect.

The first-layer squared norm is multiplied by k, each internal squared norm is unchanged, and the readout squared norm is divided by k. After including the width-dependent coefficients, the first-layer contribution is multiplied by $kC/N \le 1$, and the readout contribution by $N/(kC) < 2$. These factors both equal one if $N = kC$.

For a complexity-C witness, the original first-layer contribution is at most $d(d+1)C^2$, the internal contribution at most $(L-2)C^4$, and the readout contribution at most C^4. Applying the separate factors above gives (7). □

5 Compression by reverse-order subsampling

We now prove the converse to the cloning estimate: a bounded weighted norm gives a bounded-width implementation. The construction first approximates the real-valued output, then rescales the readout to preserve the required Boolean margin.

Theorem 5.1 (Compression). *Let $d \geq 1$, $L \geq 2$, and let $\phi : \mathbb{R} \to [-1, 1]$ be 1-Lipschitz. Suppose a width-N network W satisfies $\kappa_N(W) \leq K$, where $K \geq 1$. Set*

$$\begin{aligned} \eta &= \frac{1}{12(L-1)}, \qquad \rho = \eta 2^{-L} K^{-(L-1)/2}, \\ m_L(d, K) := M &= \max\left\{ \lceil 9K \rceil, \left\lceil 256\rho^{-4}\big(d + \log(L/\rho)\big) \right\rceil \right\}. \end{aligned} \tag{8}$$

There is a width-M network $\widetilde{W}$ such that

$$\max_{x \in \{-1,1\}^d} |z_{\widetilde{W}}(x) - z_W(x)| \leq \frac{1}{12}, \qquad \kappa_M(\widetilde{W}) \leq 4K. \tag{9}$$

If W implements a Boolean function g with strict margin $1/2$, then multiplying the readout of $\widetilde{W}$ by $3/2$ gives a width-M implementation of g with strict margin $1/2$, weighted norm at most $9K$, and every weight of absolute value at most M. In particular, $C_L(g) \leq M$. For fixed L, M is bounded by a polynomial in d and K.

Proof. We first truncate large entries and then sample the hidden neurons. All norms of rows below are Euclidean, and all matrix norms with subscript F are Frobenius norms.

Truncation. For $2 \leq s \leq L$, let $r_i^s = \|W_{i,*}^s\|_2$. In each nonzero row, delete entries larger in absolute value than

$$\tau_i^s = \frac{2r_i^s}{\rho\sqrt{N}}.$$

Leave zero rows unchanged, write $\overline{W}^s$ for the resulting matrices, and set $\overline{W}^1 = W^1$. Here and below W^L is the single readout row. Because all hidden activations have absolute value at most one, deleting a row's large entries changes its preactivation by at most

$$\sum_{j:\,|W_{ij}^s| > \tau_i^s} |W_{ij}^s| \leq \frac{(r_i^s)^2}{\tau_i^s} = \frac{\rho}{2}\sqrt{N}\, r_i^s.$$

The vector of changes at an internal layer therefore has norm at most $(\rho/2)\sqrt{N}\|W^s\|_F$. Every original or truncated internal matrix has operator norm at most $\sqrt{K}$, and the readout has norm at most $\sqrt{K/N}$. The activation is a contraction in Euclidean norm. Thus truncating layer $s < L$ changes the output by at most

$$\frac{\rho}{2}\sqrt{NK}\, K^{(L-s-1)/2}\sqrt{K/N} = \frac{\rho}{2}K^{(L-s+1)/2} \leq \frac{\eta}{2}.$$

Truncating the readout changes it by at most $(\rho/2)\sqrt{K} \leq \eta/2$. These bounds also hold when other layers have already been truncated. Telescoping over the $L-1$ truncations gives total output error at most $1/24$.

Independent samples and three events. For each hidden layer s, independently choose a list $I_s(1), \ldots, I_s(M)$ of independent uniform indices in $\{1, \ldots, N\}$. Repeated indices represent separate copies of a neuron. Define the fully sampled network by

$$\begin{aligned} \widetilde{W}_{a,*}^1 &= W_{I_1(a),*}^1, \\ \widetilde{W}_{a,b}^s &= \frac{N}{M}\overline{W}_{I_s(a),I_{s-1}(b)}^s \qquad\qquad (2 \leq s \leq L-1), \\ \widetilde{W}_b^L &= \frac{N}{M}\overline{W}_{I_{L-1}(b)}^L. \end{aligned}$$

For the analysis expose these independent lists in reverse order, $I_{L-1}, \ldots, I_1$. At the stage sampling layer m, all earlier layers still have their original, truncated weights. Let $\overline{h}_j^m(x)$ denote their exactly computed activations. The estimated sum into a retained row i of layer $m+1$ is

$$\frac{N}{M} \sum_{b=1}^{M} \overline{W}_{i,I_m(b)}^{m+1} \overline{h}_{I_m(b)}^m(x).$$

Its expectation is the corresponding full sum over N neurons. Conditional on the later lists, its summands are independent and bounded in absolute value by $(N/M)\tau_i^{m+1}$. Hoeffding's inequality [8] shows that its deviation exceeds $(\rho/2)\sqrt{N} r_i^{m+1}$ with probability at most

$$2 \exp\left(-\frac{\rho^4 M}{32}\right).$$

A zero row is estimated exactly. The bound is uniform in the retained row, so it remains valid after averaging over the later lists.

Let A be the event that all these row estimates, at all stages and all 2^d inputs, satisfy their deviation bounds. At the last hidden layer there is one output row; at every other stage there are M retained rows. Consequently

$$\Pr(A^c) \le 2^{d+1} LM \exp(-\rho^4 M/32) \le \frac{1}{100}. \tag{10}$$

For completeness, put $a = d + \log L + \log(1/\rho)$ and $M_0 = \lceil 256\rho^{-4} a \rceil$. Then $M_0 \le 257\rho^{-4}a$, and $u \exp(-\rho^4 u/32)$ decreases for $u \ge M_0$. Substituting M_0 and using $\log a \le a - 1$ bounds the logarithm of the right side before its final inequality by

$$-(7 - \log 2)d - 6 \log L - 3 \log(1/\rho) + \log 514 - 1 \le -\log 100.$$

This also proves the bound when M was enlarged to be at least $9K$.

Next define the total retained row mass using the *original* row norms:

$$R = N \sum_{s=2}^{L-1} \sum_{a=1}^{M} (r_{I_s(a)}^s)^2 + N \|W^L\|_2^2.$$

Its expectation is

$$\mathbb{E}R = M \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|W^L\|_2^2 \le MK.$$

By Markov's inequality the event $B = \{R \le 8MK\}$ has probability at least $7/8$. In particular, on B the retained row mass at each individual stage is at most $8MK$.

Finally, direct calculation gives

$$\begin{aligned} \mathbb{E}\left[\frac{d}{M}\|\widetilde{W}^1\|_F^2\right] &= \frac{d}{N}\|W^1\|_F^2, \\ \mathbb{E}\|\widetilde{W}^s\|_F^2 &= \|\overline{W}^s\|_F^2 \qquad (2 \le s \le L-1), \\ \mathbb{E}\left[M\|\widetilde{W}^L\|_2^2\right] &= N\|\overline{W}^L\|_2^2. \end{aligned}$$

For an internal matrix, independence of the two adjacent layers' index lists makes each index pair uniform on $\{1, \ldots, N\}^2$; the factor $(N/M)^2$ is canceled by the M^2/N^2 sampling factor. Independence

among different matrix entries is unnecessary. Thus $\mathbb{E}\kappa_M(\widetilde{W}) \le K$, and the event $C = \{\kappa_M(\widetilde{W}) \le 4K\}$ has probability at least $3/4$. On C every sampled internal matrix has operator norm at most $2\sqrt{K}$, and the sampled readout has norm at most $2\sqrt{K/M}$.

The events need not be independent:

$$\Pr(A \cap B \cap C) \ge 1 - \frac{1}{100} - \frac{1}{8} - \frac{1}{4} > 0.$$

Fix a collection of lists in their intersection.

Reverse-order telescoping. Let F_m be the network with hidden layers $m, \dots, L-1$ sampled and layers $1, \dots, m-1$ still original and truncated. Incoming weights to a sampled layer select its rows, and outgoing weights from a sampled layer are multiplied by N/M. This defines the intermediate networks with different widths in different layers. In particular, F_L is the truncated network and F_1 is the fully sampled network.

When comparing F_{m+1} and F_m, layer m computes the original truncated neuron functions before selection. The only new discrepancy is in the sums into the retained neurons of layer $m+1$. For $m = L-1$, event A bounds the output discrepancy directly by $(\rho/2)\sqrt{K} \le \eta/2$. For $m < L-1$, put $t = L - m$. Events A and B bound the Euclidean discrepancy in layer $m+1$'s preactivation by $(\rho/2)\sqrt{8MK}$. All matrices *after* layer $m+1$ already equal their final sampled matrices, so their operator norms are controlled by C. Propagating the discrepancy therefore changes the output by at most

$$\begin{aligned}\frac{\rho}{2}\sqrt{8MK}\,(2\sqrt{K})^{t-2}2\sqrt{K/M} &= \rho\, 2^{t-1/2}K^{t/2} \\ &\le \eta\, 2^{-3/2} \le \frac{\eta}{2}, \qquad 2 \le t \le L-1.\end{aligned}$$

There are $L-1$ sampling stages, contributing at most $1/24$ in total. Adding the truncation error proves the approximation bound in (9); event C proves its norm bound.

If the original network has strict margin $1/2$, the sampled network has strict margin $5/12$. Multiplying its linear readout by $3/2$ restores strict margin $1/2$ and multiplies the weighted norm by at most $9/4$. At width M the resulting entrywise bounds are

$$|W^1_{aj}| \le \sqrt{9KM/d}, \qquad |W^s_{ab}| \le 3\sqrt{K}\ (1 < s < L), \qquad |W^L_b| \le 3\sqrt{K/M}.$$

Each is at most M because $d \ge 1$ and $M \ge 9K \ge 9$. The definition of C_L now applies. Finally, substituting (8) gives $\rho^{-4} = 12^4(L-1)^4 16^L K^{2(L-1)}$, which establishes the stated polynomial dependence at fixed depth. □

6 Counting functions at bounded width

The compression theorem reduces the counting problem to networks of bounded width. We now count functions with low weighted-norm implementations at such a width. The argument uses the margin to separate representatives in parameter space. It does not require the posterior distribution on parameters to be supported in a bounded set.

Write $u_\ell = \sqrt{c_\ell}\, W^\ell$, where $c_1 = d/N$, $c_\ell = 1$ for $2 \le \ell \le L-1$, and $c_L = N$. Concatenating these coordinates gives $\kappa_N(w) = \|u\|_2^2$. The dimension of the parameter space is

$$P_N = N(d+1) + (L-2)N^2 + N.$$

Lemma 6.1 (Parameter sensitivity)**.** *Suppose* $|\phi| \le 1$ *and* $\mathrm{Lip}(\phi) \le 1$*. For* $d \ge 1$*,* $L \ge 2$*, and* $R > 0$*, define*

$$\Lambda_L(R,d)^2 = \frac{d+1}{d} R^{2(L-1)} + \sum_{j=0}^{L-2} R^{2j}.$$

For all rescaled parameters u, v *with* $\|u\|_2, \|v\|_2 \le R$*, and every* $x \in \{-1,1\}^d$*,*

$$|z_u(x) - z_v(x)| \le \Lambda_L(R,d)\|u - v\|_2.$$

Proof. Telescope through networks in which one layer at a time is replaced. Every layer in each intermediate network comes from either u or v, and therefore its rescaled Frobenius norm is at most R. The original first-layer Frobenius norm is at most $R\sqrt{N/d}$, each intermediate-layer norm is at most R, and the readout norm is at most $R/\sqrt{N}$.

Changing the first rescaled layer by Δu_1 changes its preactivation vector by at most $\sqrt{N/d}\sqrt{d+1}\,\|\Delta u_1\|_F$. The subsequent activations are 1-Lipschitz, so its contribution at the output is at most

$$\sqrt{\frac{d+1}{d}} R^{L-1} \|\Delta u_1\|_F.$$

For $2 \le \ell \le L-1$, the input activation vector has norm at most $\sqrt{N}$. Changing the ℓth layer therefore contributes at most

$$\sqrt{N}\,\|\Delta u_\ell\|_F R^{L-1-\ell} \frac{R}{\sqrt{N}} = R^{L-\ell}\|\Delta u_\ell\|_F$$

at the output. Changing the rescaled readout contributes at most $\|\Delta u_L\|_2$. Summing these contributions and applying Cauchy–Schwarz gives the claim. Notice that the factors of $\sqrt{N}$ cancel, so this sensitivity bound is independent of the width. □

Proposition 6.2 (Packing bound)**.** *Under the hypotheses of Lemma 6.1,*

$$\#\{g : \kappa_N(g) \le R^2\} \le \big(1 + 2R\Lambda_L(R,d)\big)^{P_N}.$$

Proof. For every counted function g, take a sequence of strict-margin implementations whose costs tend to $\kappa_N(g)$. Compactness gives a convergent subsequence in rescaled parameter space, with limit u_g satisfying $\|u_g\|_2 \le R$. Continuity of the network on the finite input domain gives $g(x)z_{u_g}(x) \ge 1/2$ for every x.

If $g \ne g'$, their labels differ somewhere, so $|z_{u_g}(x) - z_{u_{g'}}(x)| \ge 1$ for some input x. Lemma 6.1 therefore gives $\|u_g - u_{g'}\|_2 \ge 1/\Lambda_L(R,d)$. The open balls of radius $1/(2\Lambda_L(R,d))$ centered at the u_g are disjoint and lie inside the ball of radius $R + 1/(2\Lambda_L(R,d))$. Comparing their P_N-dimensional volumes proves the bound. □

7 Counting low-norm functions and proving the theorem

The preceding ingredients give both a uniform function count and a comparison of complexity measures. We record explicit quantities to make the polynomial dependence unambiguous, without attempting to optimize it.

For $K \geq 1$, use the compression width $m_L(d, K)$ from theorem 5.1, and put $m = m_L(d, K)$ and $R = 3\sqrt{K}$. Write

$$\begin{aligned} P_m &= m(d+1) + (L-2)m^2 + m, \\ \Lambda_L(R,d)^2 &= \frac{d+1}{d} R^{2(L-1)} + \sum_{j=0}^{L-2} R^{2j}, \\ Q_L(d,K) &= P_m \log(1 + 2R\Lambda_L(R,d)). \end{aligned} \tag{11}$$

For fixed L, these quantities are bounded by polynomials in d, K.

Proposition 7.1 (Uniform count and complexity comparison)**.** *For every $N \geq 1$ and $K \geq 1$,*

$$\#\{g \in \mathcal{H}_N : \kappa_N(g) \leq K\} \leq \exp Q_L(d, K+1). \tag{12}$$

Every such g has $C_L(g) \leq m_L(d, K+1)$. Together with cloning, this implies that at fixed depth, $C_L(f)$ and $\kappa_N(f)$ are polynomially related, with dependence on d and with bounds independent of N, whenever $N \geq C_L(f)$.

Proof. For each counted g, choose a strict-margin representative of norm less than $K+1$. This small slack avoids assuming that the infimum defining $\kappa_N(g)$ is attained. Compression gives a width-$m_L(d, K+1)$ representative of the same Boolean function with weighted norm at most $9(K+1)$. The packing bound gives (12).

The compression width is at least $9(K+1)$. The rescaled representative's first-layer entries have magnitude at most $3\sqrt{(K+1)m/d} \leq m$, its internal entries at most $3\sqrt{K+1} \leq m$, and its readout entries at most $3\sqrt{(K+1)/m} \leq m$. It is therefore a witness for $C_L(g) \leq m$. This implication applies to every represented g, without assuming in advance that $N \geq C_L(g)$. Cloning supplies the other direction. □

Proof of theorem 2.2. Let $A = A_{N,f}$. By cloning, $\kappa_N(f) \leq B_L(d, C)$. By Proposition 7.1,

$$\log |A| \leq Q_L(d, B_L(d, C) + 1).$$

Choose a polynomial p_L majorizing the right-hand side. By (6), there is $\sigma_0 > 0$ such that the prior mass outside A is at most $(\varepsilon/2)\pi_{N,\sigma}(f)$ whenever $0 < \sigma < \sigma_0$. Apply Lemma 3.1 with $a = \tau = \varepsilon/2$. The resulting risk bound and sample requirement are exactly those in the theorem. □

8 Boolean circuits as neural-network witnesses

This section relates the witness complexity C_L to the usual size of a Boolean circuit. We fix a finite Boolean gate basis with fan-in bounded by a constant q. Circuit depth counts gate layers, whereas network depth L counts weight matrices, including the final linear readout.

Proposition 8.1 (Simulation of a fixed Boolean gate basis)**.** *Let ϕ be continuous and nonpolynomial. Fix a finite Boolean gate basis of fan-in at most q. There is a constant a, depending only on ϕ and the basis, with the following property. If a Boolean function f has a circuit with $C \geq 1$ gates and depth D, then for every $L \geq D+1$ with $L \geq 2$, it has a depth-L network of width at most aC whose weights have absolute value at most a. The network has only the constant input $x_0 = 1$, no separate bias parameters, and its linear output equals $f(x)$ on every Boolean input. In particular, after increasing a if necessary,*

$$C_L(f) \leq aC.$$

Proof. We use the finite interpolation theorem of Pinkus [7, Theorem 5.1]: if ϕ is continuous and nonpolynomial, then arbitrary values on r distinct points can be interpolated by a linear combination of r functions of the form $v \mapsto \phi(a \cdot v - \theta)$. Apply it to the Boolean input patterns of each gate in the fixed basis, and also to the unary identity gate. Choose one representation for each:

$$G(v) = \sum_{s=1}^{r_G} \lambda_{G,s} \phi(a_{G,s} \cdot v - \theta_{G,s}), \qquad v \in \{-1, 1\}^{q_G}, \qquad r_G \le 2^{q_G}.$$

There are finitely many chosen representations, so all their coefficients are bounded by a constant depending only on ϕ and the gate basis.

We supply affine shifts using a constant unit in each hidden layer. Choose β such that $c = \phi(\beta) \neq 0$, which exists because ϕ is not the zero function. In the first layer this unit receives weight β from $x_0 = 1$. In each later hidden layer it receives weight β/c from the preceding constant unit, and again outputs c. A gate preactivation obtains its shift $-\theta$ through weight $-\theta/c$ from this constant unit (and through x_0 in the first layer).

Carry each Boolean signal as the indicated linear combination of a constant number of hidden activations. A gate in the next layer can read this signal by substituting that linear combination into its preactivation. At most q signals occur in one gate, so the resulting weights remain bounded by a constant determined by the finite dictionary. Allocate disjoint blocks of units to distinct signals; this prevents any uncontrolled accumulation of coefficients through shared encodings.

To account for edges skipping layers, relay every needed earlier signal using the identity-gate representation. A circuit with C gates and fan-in at most q references at most qC distinct input coordinates, and creates at most C gate outputs. Thus at most $(q+1)C$ Boolean signals need to be carried at any layer. Each uses a constant number of units, in addition to the one constant unit, giving width at most aC. This realizes the circuit after D hidden layers. If $L - 1 > D$, relay the final Boolean value through the remaining hidden layers by the same identity dictionary. The linear readout extracts its exact value $f(x) \in \{-1, 1\}$, which in particular satisfies the required margin. All constants used in these constructions are independent of d, C, D, L. □

Remark 8.2. Finite interpolation also shows that every Boolean function has a finite C_L for $L \ge 2$: first interpolate its values on the 2^d Boolean inputs, and then relay the resulting exact Boolean value with the identity dictionary. No approximation by near-identity layers is needed. The circuit proposition above is a forward simulation statement; it does not assert an efficient uniform procedure for evaluating an arbitrary nonpolynomial activation or for constructing the interpolants.

9 Scope and depth dependence

The argument separates two questions. The geometric question is how many Boolean functions can have low weighted norm; cloning and compression answer it uniformly in width. The probabilistic question is how much prior mass lies above a target's norm level; here we answer it only asymptotically at each fixed width. Quantitative control of this second question would be needed for a theorem at an explicitly specified nonzero variance or along a joint width–variance scaling.

Our complexity comparison is between a margin-constrained norm and a bounded-weight network implementation. It does not assert a matching lower bound on the sample complexity of learning every high-complexity target. It also does not identify the prior probability of a function with an exponential of its circuit description length: the Gaussian rate has a factor $1/(2\sigma^2)$, and equal rates need not have equal prefactors.

Depth enters compression through the amplification of errors by downstream operator norms. Proposition A.1 shows, for tanh, that low-norm sublevel sets can indeed contain doubly exponentially many functions as depth increases in an appropriate joint parameter regime. This is a limitation of a uniform norm-sublevel counting strategy. It is not a lower bound on the sample complexity of the Bayesian learner for a circuit-simple target: the competing functions would also have to receive sufficient posterior mass.

Every statistical statement concerns a fresh input from the same distribution as the training inputs, after a fixed number of independent training examples. Bounds on adversarial online mistakes or on total mistakes over an unbounded iid stream would require additional arguments.

A A depth-dependent lower bound for the norm ball

The fixed-depth restriction in the counting argument has mathematical content. For the activation $\phi = \tanh$, bounded weighted norm can encode many functions when depth grows. This is a statement about the norm ball, and does not by itself give a lower bound for learning a circuit-simple target.

Proposition A.1. *There is a universal constant A with the following property. Let $L \geq 4$, $1 \leq q \leq d$, and $N \geq 2^{q+1}$. Every Boolean function depending on a specified set of q input coordinates has a width-N, depth-L network with* tanh *hidden activations, linear readout, output exactly in $\{-1, 1\}$, and*

$$\kappa_N(w) \leq A\left[dq^2 \log^2(2q) + (L-2)2^{2q/(L-2)}\right].$$

Consequently, whenever the right-hand side is at most K, the norm ball contains at least 2^{2^q} distinct Boolean functions.

Proof. We first work at width $N_0 = 2M$, where $M = 2^q$, and write $m = L-3$. For each $p \in \{-1, 1\}^q$, introduce a first-layer unit

$$h_p(x) = \tanh(\gamma(\langle p, x\rangle - (q-1))), \qquad \gamma = \tfrac{1}{2}\log(16q),$$

where only the selected coordinates are used. The other M first-layer units are constants $b = \tanh(1)$, obtained from the constant input coordinate. The first-layer contribution to the weighted norm is exactly

$$\frac{d}{2}\left[\gamma^2(q + (q-1)^2) + 1\right] = O\Big(dq^2 \log^2(2q)\Big).$$

Index the q-cube by its vertices and define the $M \times M$ matrix $T_{xp} = (1 + h_p(x))/2$. Its diagonal differs from 1 by at most $e^{-2\gamma}$. If the Hamming distance from x to p is $r \geq 1$, then $0 < T_{xp} \leq e^{2\gamma - 4\gamma r}$. Hence

$$\|T - I\|_\infty \leq e^{-2\gamma} + e^{2\gamma}\left[(1 + e^{-4\gamma})^q - 1\right] < \tfrac{1}{4}.$$

Here $\|\cdot\|_\infty$ is the maximum absolute row sum; the last inequality follows from $e^{-2\gamma} = (16q)^{-1}$ and $(1+u)^q - 1 \leq que^{qu}$. Thus T is invertible and $\|T^{-1}\|_\infty \leq 4/3$.

Let f denote the vector of desired labels, put $v = T^{-1}f$, and fix an amplitude $0 < t_0 \leq s = 3/4$. One neuron in the second layer uses weights $a_p = \frac{1}{2}\operatorname{atanh}(t_0)v_p$ on the bump units and the common weight $(\sum_p a_p)/(Mb)$ on the constant units. Its preactivation is

$$\sum_p a_p h_p(x) + \sum_p a_p = \operatorname{atanh}(t_0) f(x),$$

so its activation is exactly $t_0 f(x)$. Its squared weight norm is $O(Mt_0^2)$, uniformly for $t_0 \leq s$. All other neurons in this layer have zero weights and activation zero.

Choose

$$t_0 = sM^{-(m-1)/(2(m+1))}, \qquad R = (s/t_0)^{1/m}, \qquad n_j = \left\lceil N_0^{j/m} \right\rceil, \quad 0 \le j \le m.$$

The jth of the next m layers has n_j active neurons, all computing $t_j f$ with $t_j = t_0 R^j$. Given the preceding n_{j-1} copies of $t_{j-1} f$, assign each incoming weight

$$\frac{\operatorname{atanh}(t_j)}{n_{j-1} t_{j-1}}.$$

This realizes the claimed activation exactly. Since $0 < t_j \le s$, $\operatorname{atanh}(t_j) \le c t_j$ for a universal constant c. Also $n_j / n_{j-1} \le 2N_0^{1/m}$. The norm contribution of this layer is therefore at most

$$\frac{n_j}{n_{j-1}} \left(\frac{\operatorname{atanh}(t_j)}{t_{j-1}} \right)^2 \le 2c^2 N_0^{1/m} R^2 = O\Big(M^{2/(m+1)}\Big).$$

The second-layer cost is of this same order because $M t_0^2 = s^2 M^{2/(m+1)}$. The last hidden layer has N_0 copies of sf; readout weights $1/(N_0 s)$ give output f and weighted readout cost s^{-2}. Summing gives the asserted bound at width N_0.

Finally, for any $N \ge N_0$, replace every hidden neuron by either $k = \lfloor N/N_0 \rfloor$ or $k+1$ identical copies, choosing the multiplicities in each layer to sum to N. Divide each outgoing weight by the multiplicity of its source neuron. This preserves the function exactly. For an inner-layer edge, its contribution to squared norm is multiplied by the ratio of its target and source multiplicities, at most $(k+1)/k \le 2$. The first- and last-layer weighted costs are likewise multiplied by at most 2, using their coefficients d/N and N. Absorbing this factor into A proves the result. □

For example, taking $q = d \ge 2$ and $L = d+2$ gives 2^{2^d} functions at weighted norm $O(d^3 \log^2(2d))$. Thus a bound on the logarithm of the norm-ball cardinality that holds uniformly in the width N and is polynomial jointly in d, K, and L is impossible. This conclusion concerns uniform counting; it does not establish that the posterior needs exponentially many samples for a particular simple target.

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